CHAPTER 2

COLLECTION OF DATA

CHAPTER ESSENCE AT A GLANCE:
Examine empirical inquiry methods: Primary versus Secondary data sources, direct personal investigation, questionnaire design rules, and Census versus Sample survey methodologies.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Primary vs Secondary Data Origin:** Primary data is original statistical material collected first-hand by the investigator for a specific purpose; Secondary data is gathered from published reports compiled by others.

• **Primary Collection Modes:** Direct Personal Interviews (high response and accuracy), Indirect Oral Investigation, Questionnaires sent through mail or digital platforms, and Schedules filled by trained enumerators.

• **Qualities of a Sound Questionnaire:** Limited number of simple questions, logical sequential order, avoidance of personal or offensive queries, clarity without mathematical calculations, and pre-testing via pilot survey.

• **Census vs Sample Investigation:** Census investigates every single item in the statistical universe (complete enumeration); Sampling investigates a representative fraction to draw inferences about the entire population.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Sampling Error Magnitude} \propto \frac{1}{\sqrt{\text{Sample Size } n}}$$

Sample Size Precision Invariant

$$\text{Total Survey Error} = \text{Sampling Errors (Chance Variation)} + \text{Non-Sampling Errors (Bias, Coverage, Response)}$$

Aggregate Statistical Survey Error Model

$$\text{Data Origin Taxonomy} = \text{Primary (Direct Empirical)} + \text{Secondary (Published and Administrative)}$$

Information Source Classification

$$\text{Representative Sample Criterion} \iff \text{Every Population Unit Has Known Non-Zero Inclusion Probability}$$

Random Probability Sampling Invariant

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A Census survey is not necessarily more accurate than a Sample survey; census operations are prone to massive non-sampling errors due to untrained enumerators.

• **Alert:** Secondary data must be scrutinized for reliability, suitability, and adequacy before using in economic research.

• **Alert:** A Pilot Survey is a pre-testing of the questionnaire on a small trial group to identify flaws and ambiguity before launching the full survey.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Primary Data and Secondary Data. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Primary Data:** Original data collected directly from the field of inquiry by the investigator for the first time. [1 Mark]

• **Secondary Data:** Data which has already been collected, tabulated, and analyzed by someone else and accessed from published or unpublished records. [1 Mark]

2. What is a Pilot Survey? State its chief advantage. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A preliminary trial survey conducted on a small sample of respondents before embarking on the large-scale investigation. [1 Mark]

• **Chief Advantage:** Helps identify confusing or ambiguous questions, assesses interview time, and tests enumerator efficiency. [1 Mark]

3. State any three essential qualities of an ideal Questionnaire. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Limited Number of Questions:** Keeps the questionnaire brief and focused strictly on the inquiry objectives to prevent respondent fatigue. [1 Mark]

• **Simplicity and Clarity:** Questions must be clear, straightforward, and easily understandable without complex mathematical calculations. [1 Mark]

• **Logical Sequence:** Questions should proceed logically from general background queries to specific technical topics. [1 Mark]

4. Distinguish between the Census Method and the Sampling Method of data collection. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Coverage:** Census examines every individual element in the universe; Sampling examines only a representative cross-section. [1 Mark]

• **Cost and Time:** Census involves enormous financial expenditure, extensive manpower, and lengthy time; Sampling is economical, rapid, and requires fewer investigators. [1 Mark]

• **Destructive Testing:** Census is impossible when items are destroyed during inspection (e.g. testing bullet quality or blood samples); Sampling is compulsory. [1 Mark]

5. Distinguish between Sampling Errors and Non-Sampling Errors. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Sampling Errors:** Errors arising due to the chance fact that only a sample rather than the complete population is observed (decreases as sample size increases). [1.5 Marks]

• **Non-Sampling Errors:** Errors arising from human mistakes, such as faulty measurement, bias in interview, non-response, or recording errors (present in both sample and census surveys). [1.5 Marks]

6. State two major national-level organizations in India that collect and publish secondary statistical data. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **National Statistical Office (NSO):** Conducts nationwide socio-economic sample surveys on employment, consumer expenditure, and unorganized enterprises under MoSPI. [1.5 Marks]

• **Registrar General of India (RGI):** Conducts the decennial population census of India and records vital statistics (birth and death rates). [1.5 Marks]

7. Explain the major methods of collecting Primary Data: Direct Personal Investigation, Mailed Questionnaire, and Schedules filled by Enumerators. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **Direct Personal Investigation:** Investigator personally interviews respondents; yields highly accurate, reliable, and in-depth responses but is costly and restricted in geographical coverage. **[1.5 Marks]**

• **Mailed Questionnaire Method:** Questionnaires mailed or emailed with request to fill and return; covers wide geographic areas at low cost but suffers from high non-response and misinterpretation. **[1.25 Marks]**

• **Schedules Sent Through Enumerators:** Trained enumerators visit respondents, ask questions, and record answers on schedules; combines wide coverage with high accuracy and low non-response. **[1.25 Marks]**